Then, $f(2a-x) = f(2\pi - x) = g(\cos(2\pi - x)) = g(\cos x) = f(x)$

$$\therefore \int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx .$$

$$\therefore \int_0^{2\pi} g(\cos x)\,dx = 2\int_0^{\pi} g(\cos x)\,dx .$$

Result

$$\int_0^{2\pi} g(\cos x)\,dx = 2\int_0^{\pi} g(\cos x)\,dx.$$

Note

The above result is useful in evaluating definite integrals of the type $\int_0^{2\pi} g(\cos x)\,dx.$

Example 9.23

If $f(x) = f(a+x)$, then $\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$

Solution

We write $\int_0^{2a} f(x)\,dx = \int_0^{a} f(x)\,dx + \int_a^{2a} f(x)\,dx$ ... (1)

Consider $\int_a^{2a} f(x)\,dx$

Substituting $x = a+u$, we have $dx = du$; when $x = a, u = 0$ and when $x = 2a, u = a$.

$$\therefore \int_a^{2a} f(x)\,dx = \int_0^{a} f(a+u)\,du = \int_0^{a} f(u)\,du, \text{ since } f(x) = f(a+x)$$

$$= \int_0^{a} f(x)\,dx . \qquad \text{... (2)}$$

Substituting (2) in (1), we get

$$\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx .$$

Example 9.24

Evaluate : $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\cos x\,dx.$

Solution

Let $f(x) = x\cos x$. Then $f(-x) = (-x)\cos(-x) = -x\cos x = -f(x).$

So $f(x) = x\cos x$ is an odd function.

Hence, applying the property, for odd function *f(x)*, $\int_{-a}^{a} f(x)dx = 0$,

$\therefore$ we get $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\cos x\,dx = 0.$

Example 9.25

Evaluate : $\int_{-\log 2}^{\log 2} e^{-|x|}\,dx$.

Solution

Let $f(x) = e^{-|x|}$. Then $f(-x) = e^{-|-x|} = e^{-|x|} = f(x)$

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